Course Notes: Pearson Edexcel International A Level Physics

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1 Physics Year 1

Textbook page 56

Consider a constant force $\vec{F}$ causing a ball of mass m which is moving at a velocity $\vec{v}$ to come to rest over a distance $\vec{s}$.

TODO: TikZ diagram

Using SUVAT:

$$v^2 = u^2 + 2as \tag{1}$$

But since it comes to rest, $v = 0$, and the initial velocity $u = |\vec{v}|$. Rearranging for a:

$$\begin{aligned} 0 &= v^2 + 2as \\ v^2 &= -2as \\ -\frac{v^2}{2} &= as \\ as &= -\frac{v^2}{2} \end{aligned} \tag{2}$$

Since the work done W by a constant force is given by:

$$W = \vec{F} \cdot \vec{s} = m\vec{a} \cdot \vec{s} \tag{3}$$

We can substitute equation 2 into this to get something already resembling the kinetic energy formula:

$$\begin{aligned} W &= m \cdot -\frac{v^2}{2} \\ W &= -\frac{1}{2}mv^2 \end{aligned}$$

Since the work done W is equal to the change in kinetic energy, and the particle ends at rest, we have:

$$\begin{aligned} W &= E_{kf} - E_{ki} \\ W &= 0 - E_{ki} \\ W &= -E_{ki} \\ E_{ki} &= -W \\ E_{ki} &= \frac{1}{2}mv^2 \end{aligned}$$

6 Electric and Magnetic Fields

6C Electromagnetic Effects

Gravitational, electric and magnetic fields are all vector fields. The strength of each field is defined as the force per unit something experienced by an object placed in the field:

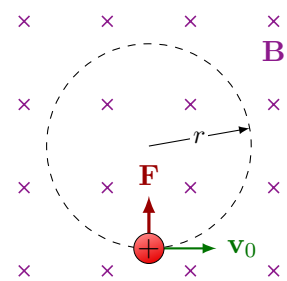
$$\vec{g} = \frac{\vec{F}}{m} \qquad \vec{E} = \frac{\vec{F}}{q} \qquad \vec{B} = \frac{\vec{F}}{q\vec{v}}$$

$$\Phi = \frac{W}{m} \qquad V = \frac{W}{q} \qquad ? = \frac{W}{q\vec{v}}$$

Or V_{grav}

$$\begin{aligned} W &= \vec{F} \cdot \vec{s} \\ W &= |\vec{F}|\,|\vec{s}|\,\cos\theta \end{aligned}$$

Magnetic fields can cause circular motion!



The forces involved are the magnetic force and the centripetal force:

$$\vec{F}_B = q\vec{v} \cdot \vec{B} \qquad \vec{F}_{cr} = \frac{m\vec{v}^2}{r}$$

Equating the magnetic force to the centripetal force, we can solve for the radius of the circular path:

$$
\begin{aligned}
q\vec{v} \cdot \vec{B} &= \frac{m\vec{v}^2}{r} \\
q\vec{B} &= \frac{m\vec{v}}{r} \\
r &= \frac{m\vec{v}}{q\vec{B}} \\
r &= \frac{\vec{p}}{q\vec{B}}
\end{aligned}
$$

See also

- Raycast AI Chat, Gemini 2.5 Pro: Electromagnetic Induction and Magnetic Forces: Equations, hints, and problem-solving

Magnetic flux through an area A in a magnetic field B is given by:

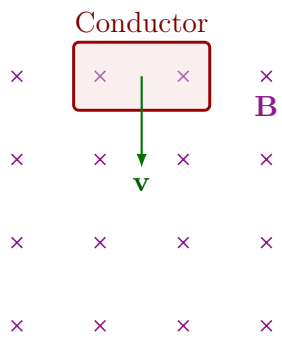
$$
\begin{aligned}
\Phi &= \vec{B} \cdot \vec{A} \\
\Phi &= |\vec{B}|\,|\vec{A}|\,\cos\theta
\end{aligned}
$$

Where Φ is in webers, Wb, or $\mathrm{T\,m^2}$, since B is in teslas and A is in square metres.

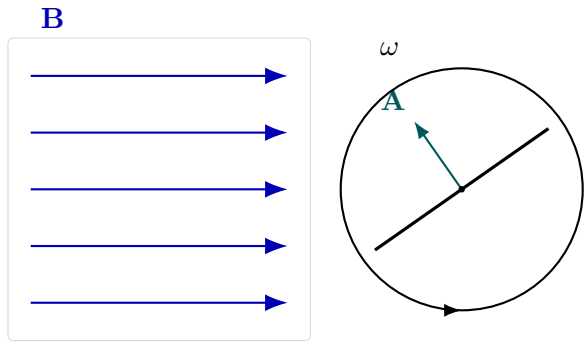
More magnetic flux stuff:

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\Phi &= V \\
\frac{\mathrm{d}}{\mathrm{d}t}N\Phi &= V
\end{aligned}
$$

"The rate of change of flux linkage through a coil gives the induced e.m.f. across the coil."



Or we can have induction with a rotating coil in a magnetic field:



Here, e.m.f. is given by:

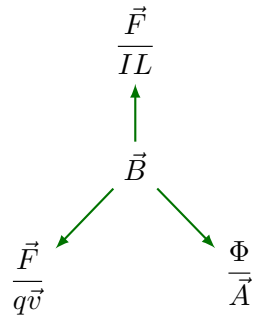
$$
\mathcal{E} = \frac{\mathrm{d}}{\mathrm{d}t}[N\Phi] \tag{4}
$$

Taking out the constants, rearranging and substituting:

$$
\begin{aligned}
\mathcal{E} &= N\frac{\mathrm{d}\Phi}{\mathrm{d}t} \\
\mathcal{E} &= N\frac{\mathrm{d}}{\mathrm{d}t}\left(\vec{B} \cdot \vec{A}\right) \\
\mathcal{E} &= N\frac{\mathrm{d}}{\mathrm{d}t}\left(|\vec{B}|\,|\vec{A}|\,\cos\theta\right)
\end{aligned}
$$

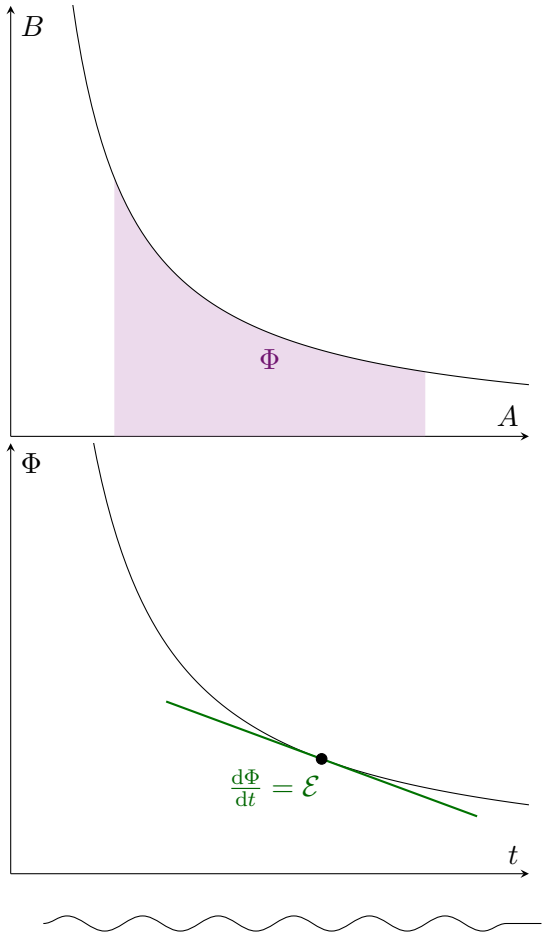
Using $A = |\vec{A}|$ and $B = |\vec{B}|$ for simplicity:

$$
\begin{aligned}
\mathcal{E} &= N\frac{\mathrm{d}}{\mathrm{d}t}(BA\cos\theta) \\
\mathcal{E} &= NBA\frac{\mathrm{d}}{\mathrm{d}t}[\cos\theta] \\
\mathcal{E} &= NBA\frac{\mathrm{d}}{\mathrm{d}t}[\cos\omega t] \\
\mathcal{E} &= -NBA\omega\sin(\omega t)
\end{aligned}
$$



Therefore,

$$\vec{F} = \vec{B}IL$$
$$\vec{F} = q\vec{v} \cdot \vec{B}$$

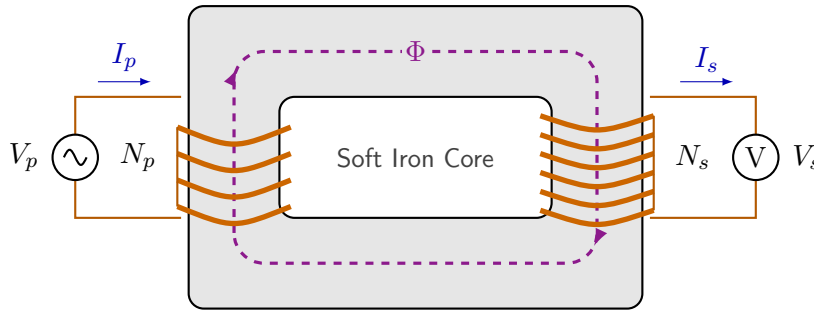


TRANSFORMERS!!!

Ratio of primary to secondary e.m.f.s is equal to the ratio of the number of turns in the primary to the secondary coil.

$$\frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{N_p}{N_s} \tag{5}$$

This equation can be derived using the following diagram of a transformer and the definition of e.m.f. seen in equation 4:



From the diagram, we can see that the magnetic flux Φ through each coil is the same. Therefore, we can write the e.m.f.s in each coil as:

$$\mathcal{E}_p = -N_p \frac{d\Phi}{dt}, \quad \mathcal{E}_s = -N_s \frac{d\Phi}{dt} \tag{6}$$

Taking the ratio of these two equations gives us equation 5:

$$\frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{N_p}{N_s} \tag{5 revisited}$$

7 Nuclear and Particle Physics

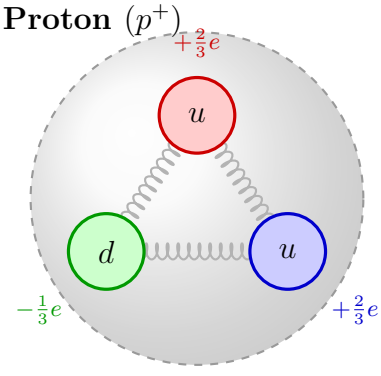
7C The Particle Zoo

Textbook page 100

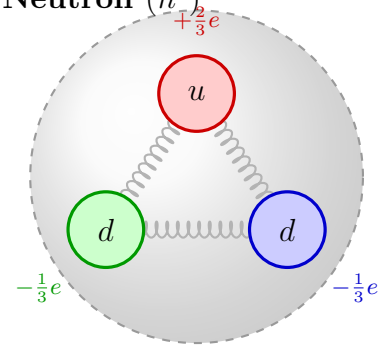
"Hadrons are subatomic particles made of quarks, which experience a Strong Nuclear Force."

"Quark flavour changing is done by the Weak Nuclear Force."

Proton (p^+)



Neutron (n^0)



A List of Equations

AA Waves

- $v = \lambda f$
- $v^2 = \frac{T}{\mu}$ in a uniform string/rope

  Force of tension

  Linear density in $\mathrm{kg\,m^{-1}}$

Interference:

- p.d. $= n\lambda$ ← constructive
- p.d. $= \left(n + \frac{1}{2}\right)\lambda$ ← destructive

AB Photoelectric Effect

- $h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
- $E = hf$
- $hf = \phi + E_{k\mathrm{max}}$
- $V = \frac{W}{Q} \iff W = QV$

B List of Definitions

BA Circuits

e.m.f.

- Energy supplied per unit charge
- Work done per unit charge
- The work done moving unit charge around the whole circuit

C Textbook

The textbook pages are omitted in draft mode.